

Assignment 2 Report

Building Detection from Satellite Imagery

1. Problem

What are we building and why is it useful?

We are building an automated object detection pipeline that takes satellite imagery as input and automatically detects and counts buildings. This is highly useful for rapid urban planning, disaster response damage assessment, infrastructure mapping in remote areas, and population density estimation.

2. Data

Where it came from: The dataset is the SpaceNet 2 Paris Buildings dataset, downloaded via Kaggle ([ugorjiir/spacenet-2-paris-buildings](https://www.kaggle.com/ugorjiir/spacenet-2-paris-buildings)).

How many images: There are a total of 1,529 images. The dataset was split into 918 for training, 230 for validation, and 381 for testing.

What the labels look like: The original labels were in GeoJSON format containing geographic coordinates (longitude/latitude). These were converted into YOLO .txt format, where each line represents a building bounding box formatted as: class_id cx cy bw bh (where the coordinates are normalized pixel values between 0 and 1).

3. MODEL

- What is YOLOv8?: YOLOv8 (You Only Look Once, version 8) is a state-of-the-art, real-time object detection model built by Ultralytics. It is designed to be highly fast and accurate for identifying objects in images.
- Which size did you pick: YOLOv8s (Small variant) using Transfer Learning.
- How many parameters: The model has 11,125,971 parameters (approx. 11.1 million).

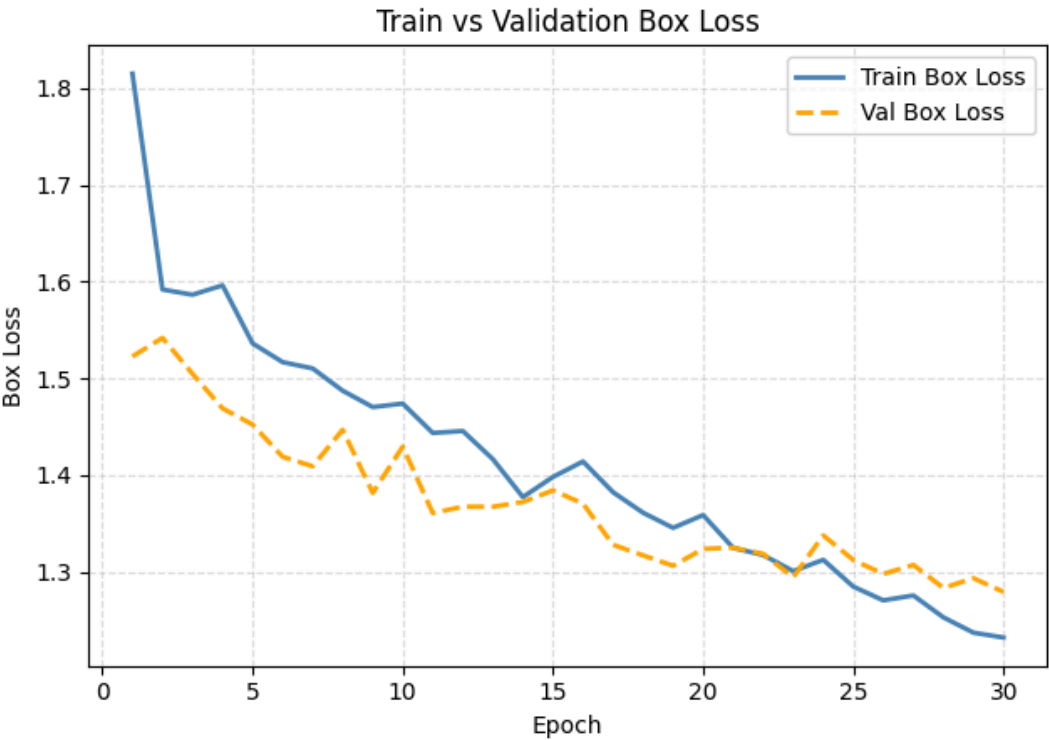
4. TRAINING

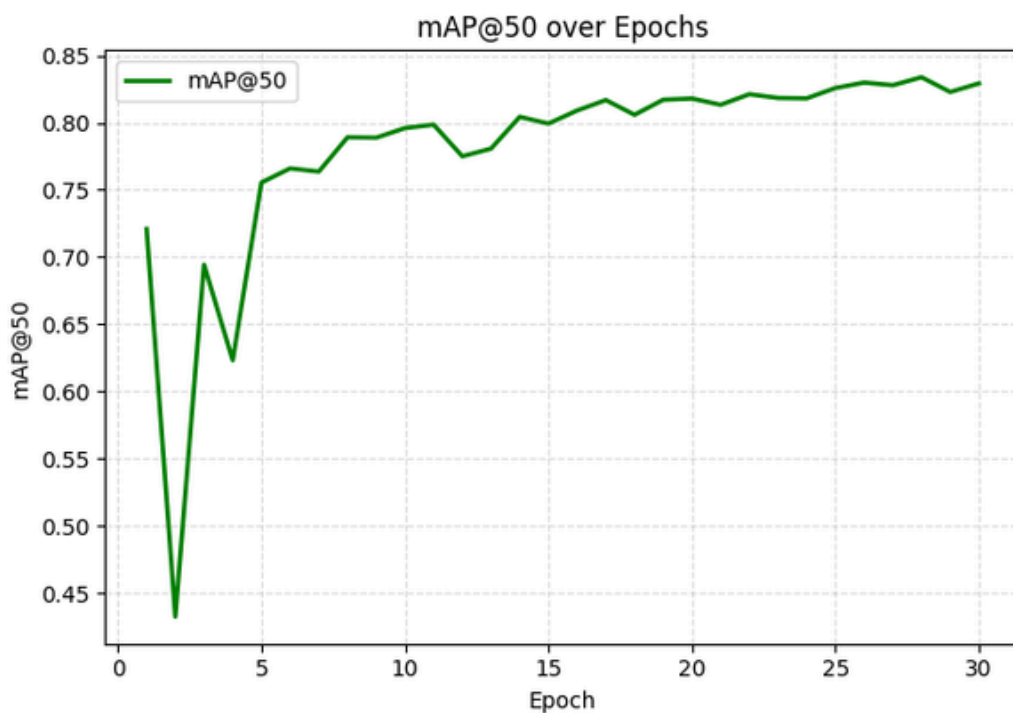
- Epochs: 30 epochs
- Batch size: 16
- GPU used: Tesla T4 (Google Colab)
- Training time: Roughly 18.5 seconds per epoch (Total training time was approximately 9.5 minutes).

5. RESULTS

Here are your final metrics from Epoch 30, including the calculated F1 score:

Metric	Your Value	What it means in plain English
mAP@50	0.8292	Mean Average Precision. Overall, the model is ~83% accurate at drawing correct bounding boxes around buildings.
Precision	0.8379	Out of all the objects the model <i>claimed</i> were buildings, 83.79% of them were actually buildings (low false positives).
Recall	0.7610	Out of all the <i>actual</i> buildings in the dataset, the model successfully found 76.10% of them (missed roughly 24%).
F1 Score	0.7976	The harmonic mean of Precision and Recall. It shows the model has a very strong balanced performance of ~80%.





6. OBSERVATIONS

What worked well: The model performed excellently on images with sparse or no buildings (perfectly predicting 0 buildings when none were present, and 2 out of 2 when isolated). The overall Mean Absolute Error (MAE) for counting was very low at just 2.90 buildings per image.

What the model struggled with: Based on the test loop, the model struggled slightly with densely packed urban areas, tending to over-count in clusters (e.g., predicting 46 buildings when there were 28, or 66 when there were 59). This might be due to shadows or overlapping rooftops.

One idea to improve it: To improve performance on dense clusters, you could increase the training epochs (e.g., to 50 or 100), apply heavier data augmentation (like mosaic or rotation) to teach the model to handle weird angles, or upgrade to a larger model like YOLOv8m (Medium).